

May 13

Analytic tools — Hamilton's 3-manifold theorem

Philosophy Ricci tensor = Laplacian of metric

(rough) Laplacian acting on functions:

$$u \text{ function: } \Delta u = \text{trace}(\underbrace{\nabla^2 u}_{\text{symmetric 2-tensor}})$$

or tensor

$$(M^n, g) \quad x \in M, r > 0,$$

$$\int_{B_r(x)} u = u(x) + \frac{1}{2(n+2)} \Delta u(x) \cdot r^2 + O(r^4).$$

Laplacian represents averaging

RF: smoothing.

$$M^n, g(t) : \frac{\partial}{\partial t} g = -2 \text{Ric}. \quad \text{RF}$$

Recall:

$$\frac{\partial R}{\partial t} = \Delta R + 2|\text{Ric}|^2$$

parabolic maximum principle:

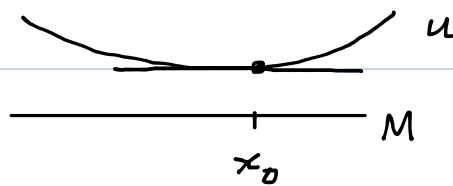
= 1st & 2nd derivative tests from multivariable calculus.

Suppose $u: M \rightarrow \mathbb{R}$ attains its minimum

at $x_0 \in M$.

1st deriv. test: $\nabla u(x_0) = 0$.

2nd deriv test: $\nabla^2 u(x_0) \geq 0$.



Lemma M^n closed, $g(t)$ 1-param. fam.

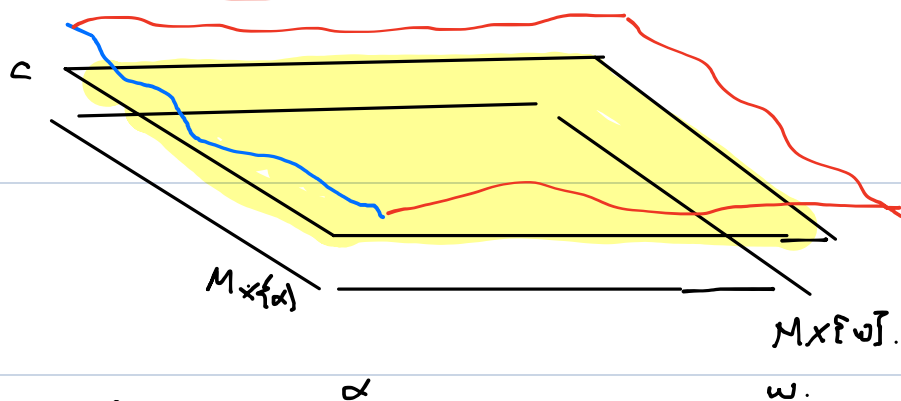
Suppose $u: M \times [\alpha, \omega) \rightarrow \mathbb{R}$ satisfies:

$$\partial_t u \geq \Delta u + X(t) \cdot \nabla u$$

where $X(t)$ is a time-dep. vector field on M .

If $u \geq c$ on M at $t = \alpha$, then.

$$u \geq c \text{ on } M \times [\alpha, \omega).$$



This is an a priori estimate:

Gives important control on the solution.

Corollary Under RF on $[0, T)$, if

$R \geq c$ at $t = 0$, then $R \geq c$ on $[0, T)$.

Proof Apply max. prin. to $\partial_t R = \Delta R + 2|R|c \geq \Delta R$.

□

Proof of parabolic max. prin.

① Idea: Suppose at $t_0 \in [\alpha, \omega)$,

u on M attains a min. at x_0 .

Then $\nabla u(x_0, t_0) = 0$, $\Delta u(x_0, t_0) \geq 0$.

So $\partial_t u(x_0, t_0) \geq 0$.

② Rigorous proof: Let $\varepsilon > 0$ & $\varepsilon \rightarrow 0$.

Def $u_\varepsilon = u + \varepsilon(t + \varepsilon)$.

Calc. eqn for u_ε & apply maximum principle

Answer: Get $u_\varepsilon \geq c$ on $M \times [\alpha, \omega)$. Then $\varepsilon \rightarrow 0$. $\square$

α 2-tensor: $|\alpha|^2 \geq \frac{1}{n} \text{trace}(\alpha)^2$

so $|\text{Ric}|^2 \geq \frac{1}{n} R^2$.

So $\boxed{\partial_t R = \Delta R + 2|\text{Ric}|^2 \geq \Delta R + \underbrace{\frac{2}{n} R^2}_{\text{fcn of } R}}.$

Parab. max prin, II. As before. Now:

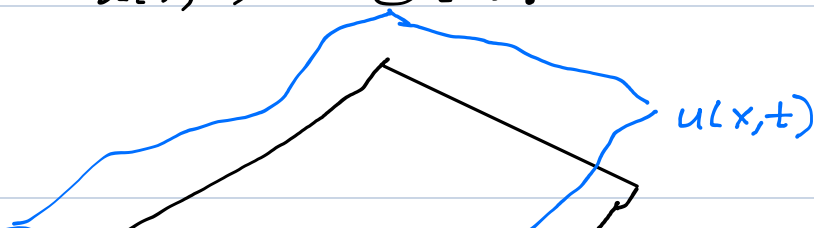
$$\partial_t u \geq \Delta u + X(t) \cdot \nabla u + F(u).$$

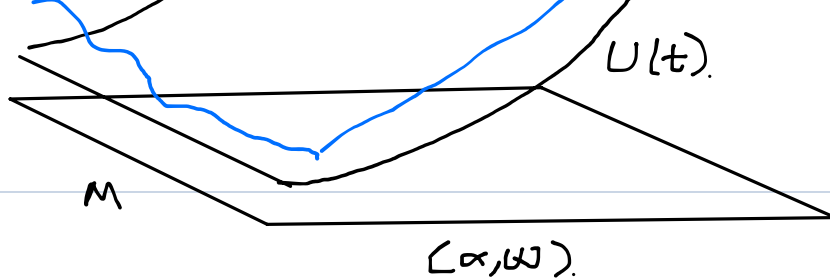
Let $U(t)$ be the solution to the ODE.

$$\frac{dU}{dt} = F(U), \quad U(0) = c.$$

If $u \geq c$ at $t = \alpha$, then.

$$u(x, t) \geq U(t).$$





Corollary Under FF, if $R \geq s_0$ at $t=0$,

then $R(x, t) \geq \frac{n s_0}{n - 2 s_0 t} =: s(t)$.

$$\uparrow \frac{ds}{dt} = \frac{2}{n} s^2, \quad s(0) = s_0$$

Analogy: heat eqn.

$$\partial_t u = \Delta u.$$

$$\partial_t |\nabla u|^2 = \Delta |\nabla u|^2 - 2 |\nabla^2 u|^2 - 2 \text{Ric}(\nabla u, \nabla u).$$

u solution, $|\nabla u|^2$ gradient quantity.

get a gradient estimate.

Morally. $\partial_t g = \text{"}\Delta\text{"} g.$

Dirichlet energy. $\int_M |\partial g|^2$ depends on coord.
nothing like this makes sense tensorially.

Need to be careful making analogies.

Under RF,
Q: heat eqn for Ric?

It's important to know what to calculate.

From last time.

$$\begin{aligned}
 \frac{\partial}{\partial t} R_{jk} &= -g^{il} \left(\underbrace{\nabla_i \nabla_j R_{kl}}_{\text{blue}} + \underbrace{\nabla_i \nabla_k R_{jl}}_{\text{blue}} - \underbrace{\nabla_i \nabla_l R_{jk}}_{\text{yellow}} - \underbrace{\nabla_j \nabla_k R_{il}}_{\text{green}} \right) \\
 &= \Delta R_{jk} + \underbrace{\nabla_j \nabla_k R}_{\text{blue}} \\
 &\quad - \underbrace{g^{il} \nabla_j \nabla_i R_{kl}}_{\text{yellow}} + g^{il} \left(\underbrace{R_{ijkm} R_{ml}}_{\text{blue}} + \underbrace{R_{ijlm} R_{km}}_{\text{green}} \right) \\
 &\quad - \underbrace{g^{il} \nabla_k \nabla_i R_{jl}}_{\text{yellow}} + g^{il} \left(\underbrace{R_{ikjm} R_{ml}}_{\text{blue}} + \underbrace{R_{iklm} R_{jm}}_{\text{green}} \right) \\
 \boxed{\partial_t R_{jk} = \Delta R_{jk} + 2g^{il} R_{ijkm} R_{ml} - 2R_{jm} R_{km}}
 \end{aligned}$$

since $g^{il} \nabla_i R_{kl} = \frac{1}{2} \nabla_k R$

Sign convention: on S^2 : $R_{1221} = 1$.

Lichnerowicz Laplacian.

def. h symm 2-tensor:

$$\begin{aligned}
 \Delta_L h_{jk} &= \Delta h_{jk} + 2g^{il} R_{ijkm} h_{ml} \\
 &\quad - R_{jm} h_{km} - R_{km} h_{jm}.
 \end{aligned}$$

So $(\partial_t \text{Ric})_{jk} = (\Delta_L \text{Ric})_{jk}$.

The "problem of Rm " was solved by Hamilton in 3D:

$$\begin{aligned}
 R_{ijkl} &= R_{il} g_{jk} + R_{jk} g_{il} - R_{ik} g_{jl} - R_{jl} g_{ik} \\
 &\quad - \frac{1}{2} R (g_{il} g_{jk} - g_{ik} g_{jl}).
 \end{aligned}$$

A brt algebra $\Rightarrow$

$$\partial_t R_{ij} = \Delta R_{ij} + Q_{ij}.$$

where

$$Q_{ij} = 3R R_{ij} - 6 R_{il} R_{jl} + (2|Ric|^2 - R^2) g_{ij}.$$

$Ric \geq 0$ means $Ric(V, V) \geq 0 \quad \forall V$.

i.e. (R_{ij}) positive semi-definite symm. matrix.

In RF, we want to control the geometry
of $g(t)$.

For example. control curvature.

Lemma. In dimension 3, under RF on a closed manifold M^3 on $[0, T)$, if $Ric \geq 0$ at $t=0$, then $Ric \geq 0$ for all $t \in [0, T)$.

Under RF

$$(3D) \quad \partial_t Ric = \Delta Ric + Q.$$

"quad. in Ric."

Hamilton's tensor maximum principle:

Under RF, 3D, since

$$\partial_t Ric = \Delta Ric + Q.$$

satisfies the following. null-eigenvector condition

If $Ric(V) = 0$ at some (x_0, t_0) ,
then $Q(V, V) \geq 0$ at (x_0, t_0) .

If $Ric \geq 0$ at $t=0$, then $Ric \geq 0$ for $t \in [0, T)$.

Some amount of geometric control on $g(t)$.

We want to improve the control.

to the extent we eventually
get convergence to an Einstein metric

$\text{Ric} \equiv \frac{1}{3} R g$ after rescaling.